

Applied Mathematics for Chemists

(화학수학)

Lecture 13

Differential Equations



Linear ODEs with Constant Coefficients

- Homogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Inhomogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

Inhomogeneous Linear ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

1. Solve the complementary equation →
complementary function (보상함수) y_c
2. Find a particular solution y_p that satisfies the DE
3. The general solution is $y = y_c + y_p$

Guess for Particular Solution

Inhomogeneous term	Trial solution
1	A
t^n	$A_0 + A_1 t + A_2 t + \cdots + A_n t^n$
$e^{\alpha t}$	$Ae^{\alpha t}$
$t^n e^{\alpha t}$	$e^{\alpha t}(A_0 + A_1 t + A_2 t + \cdots + A_n t^n)$
$e^{\alpha t} \sin(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$
$e^{\alpha t} \cos(\beta t)$	$e^{\alpha t}[A \cos(\beta t) + B \sin(\beta t)]$

Glitch in the Method

A trial solution y_p can be
a solution of the complementary equation.

In this case, try $x^n y_p$ instead!

(n : smallest integer that resolves the problem)

Application: Forced Harmonic Oscillator

$$F = m \frac{d^2 x}{dt^2} = -kx + \underbrace{F(t)}_{\text{external force}}$$

- Standard form

$$\frac{d^2 x}{dt^2} + \frac{k}{m} x = \frac{F(t)}{m}$$

- Complementary equation

$$\frac{d^2 x}{dt^2} + \frac{k}{m} x = 0$$

$$\rightarrow x_c(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

Laplace Transforms

- Laplace transform

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt = \mathcal{L}\{f(t)\}$$

- Inverse Laplace transform

$$f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)e^{st} ds = \mathcal{L}^{-1}\{F(s)\}$$

requires complex analysis;
We will not use this formula

Some Common Laplace Transforms

$F(s)$	$f(t)$
$1/s$	1
$1/s^2$	t
$n!/s^{n+1}$	t^n
$\frac{1}{s-a}$	$e^{at} (s > a)$
$\frac{s}{s^2 + a^2}$	$\cos(at)$
$\frac{a}{s^2 + a^2}$	$\sin(at)$
$\frac{s}{s^2 - a^2}$	$\cosh(at)$
$\frac{a}{s^2 - a^2}$	$\sinh(at)$

Useful Theorems

- Shifting Theorem

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

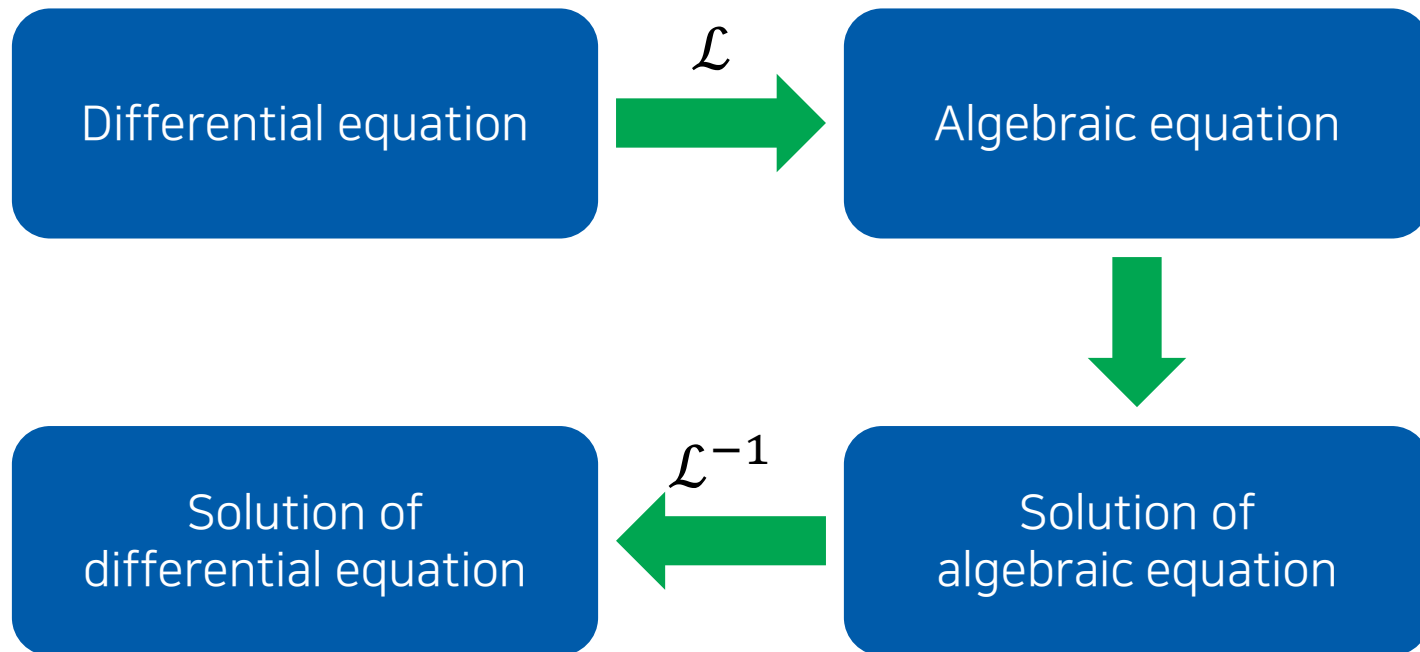
- Derivative theorem

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\mathcal{L}\{f(t)\} - f(0)$$

- Integral theorem

$$\mathcal{L}\left\{\int_0^t f(u) du\right\} = \frac{1}{s}\mathcal{L}\{f(t)\}$$

Solving DE with Laplace Transforms



Today's Class

- Partial differential equations
 - Separation of variables
 - Wave equation
 - Schrödinger equation

Partial Differential Equations

- General form of a linear second-order PDE

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

where $A, B, \dots, G$ are functions of x and y .

- $G(x, y) = 0$: homogeneous
- $G(x, y) \neq 0$: inhomogeneous

Separation of Variables

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

Assume that $u(x, y) = X(x)Y(y)$.

$$\begin{aligned} X' &= dX/dx, X'' = d^2X/dx^2, \dots \\ Y' &= dY/dy, Y'' = d^2Y/dy^2, \dots \end{aligned}$$

If we obtain $f(x, X, X', \dots) = g(y, Y, Y', \dots)$, let

$$f(x, X, X', \dots) = g(y, Y, Y', \dots) = \lambda$$

where λ is the **separation constant**.

(Tip: it is often convenient to use $\lambda = \kappa^2$ or $-\kappa^2$)

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial u}{\partial y}.$$

Assume $u(x, y) = X(x)Y(y)$ and obtain

$$X''Y = 4XY'$$

$$\frac{X''}{4X} = \frac{Y'}{Y}$$

Each side must be a constant. Consider the following three cases:

- 1) Separation constant > 0
- 2) Separation constant < 0
- 3) Separation constant $= 0$

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial u}{\partial y}.$$

1) Case 1:

$$\frac{X''}{4X} = \frac{Y'}{Y} = \kappa^2$$

Hence,

$$X'' - 4\kappa^2 X = 0, \quad Y' - \kappa^2 Y = 0$$

which have the solutions

$$X(x) = c_1 e^{2\kappa x} + c_2 e^{-2\kappa x}, \quad Y(y) = c_3 e^{\kappa^2 y}$$

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial u}{\partial y}.$$

1) Case 1:

$$X(x) = c_1 e^{2\kappa x} + c_2 e^{-2\kappa x}, \quad Y(y) = c_3 e^{\kappa^2 y}$$

The final solution is

$$u(x, y) = X(x)Y(y) = (c_1 e^{2\kappa x} + c_2 e^{-2\kappa x})c_3 e^{\kappa^2 y}$$

or

$$u(x, y) = A_1 e^{2\kappa x + \kappa^2 y} + B_1 e^{-2\kappa x + \kappa^2 y}$$

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial u}{\partial y}.$$

2) Case 2:

$$\frac{X''}{4X} = \frac{Y'}{Y} = -\kappa^2$$

Hence,

$$X'' + 4\kappa^2 X = 0, \quad Y' + \kappa^2 Y = 0$$

which have the solutions

$$X(x) = c_4 \cos 2\kappa x + c_5 \sin 2\kappa x, \quad Y(y) = c_6 e^{-\kappa^2 y}$$

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial u}{\partial y}.$$

2) Case 2:

$$X(x) = c_4 \cos 2\kappa x + c_5 \sin 2\kappa x, \quad Y(y) = c_6 e^{-\kappa^2 y}$$

The final solution is

$$u(x, y) = X(x)Y(y) = (c_4 \cos 2\kappa x + c_5 \sin 2\kappa x)c_6 e^{-\kappa^2 y}$$

or

$$u(x, y) = A_2 e^{-\kappa^2 y} \cos 2\kappa x + B_2 e^{-\kappa^2 y} \sin 2\kappa x$$

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial u}{\partial y}.$$

3) Case 3:

$$\frac{X''}{4X} = \frac{Y'}{Y} = 0$$

Hence,

$$X'' = Y' = 0$$

which have the solutions

$$X(x) = c_7 x + c_8, \quad Y(y) = c_9$$

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial u}{\partial y}.$$

3) Case 3:

$$X(x) = c_7 x + c_8, \quad Y(y) = c_9$$

The final solution is

$$u(x, y) = X(x)Y(y) = (c_7 x + c_8)c_9$$

or

$$u(x, y) = A_3 x + B_3$$

Example

Find solutions of

$$\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial u}{\partial y}.$$

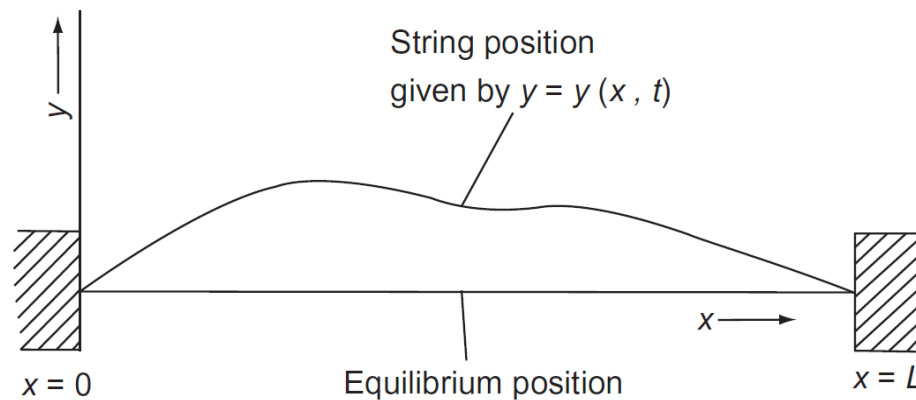
- 1) Separation constant > 0 : $u(x, y) = A_1 e^{2\kappa x + \kappa^2 y} + B_1 e^{-2\kappa x + \kappa^2 y}$
- 2) Separation constant < 0 : $u(x, y) = A_2 e^{-\kappa^2 y} \cos 2\kappa x + B_2 e^{-\kappa^2 y} \sin 2\kappa x$
- 3) Separation constant $= 0$: $u(x, y) = A_3 x + B_3$

All solutions satisfy the differential equation, and the general solution is

$u(x, y)$

$$= A_1 e^{2\kappa x + \kappa^2 y} + B_1 e^{-2\kappa x + \kappa^2 y} + A_2 e^{-\kappa^2 y} \cos 2\kappa x + B_2 e^{-\kappa^2 y} \sin 2\kappa x + A_3 x + B_3$$

Waves in a String



- Assumptions
 1. The string is perfectly flexible and homogeneous
 2. The displacements y are small compared to L
 3. Each segment of the string follows Hooke's law

Wave Equation (파동방정식)

From the equation of motion (Newton's law),

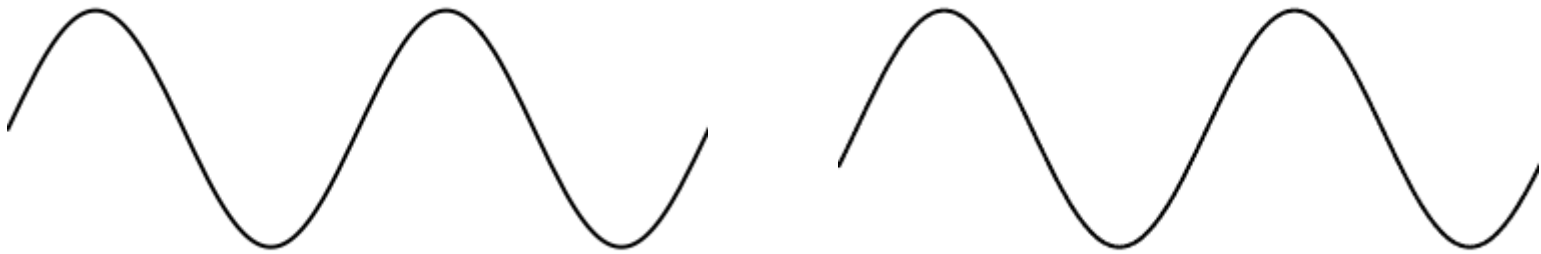
$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\rho} \frac{\partial^2 y}{\partial x^2}$$

where T is the magnitude of the tension force and ρ is the mass of the string per unit length.

Letting $c = \sqrt{T/\rho}$,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Traveling Waves (진행파)



$$y(x, t) = A \sin[k(x - ct)]$$

Example 12.12

Show that the function $y(x, t) = A \sin[k(x - ct)]$ satisfies the wave equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}.$$

The left-handed side is

$$\frac{\partial^2 y}{\partial t^2} = -Ak^2 c^2 \sin[k(x - ct)]$$

and the right-handed side is

$$c^2 \frac{\partial^2 y}{\partial x^2} = -Ak^2 c^2 \sin[k(x - ct)]$$

Hence, the function satisfies the wave equation.



Another Case: Additional Conditions

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

- Boundary conditions

$$y(0, t) = y(L, t) = 0$$

- Initial conditions

$$y(x, 0) = 0, \quad \left. \frac{\partial y}{\partial t} \right|_{t=0} = f(x)$$

Solving Wave Equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Assume $y(x, t) = \psi(x)\theta(t)$;

$$\psi \frac{d^2 \theta}{dt^2} = c^2 \frac{d^2 \psi}{dx^2} \theta$$

$$\underbrace{\frac{1}{c^2 \theta} \frac{d^2 \theta}{dt^2}}_{\text{temporal part}} = \underbrace{\frac{1}{\psi} \frac{d^2 \psi}{dx^2}}_{\text{spatial part}} = \lambda$$

Spatial Part

$$\frac{d^2\psi}{dx^2} = \lambda\psi$$

1. If $\lambda > 0$, we can set $\lambda = \kappa^2$ and obtain

$$\psi(x) = a_1 e^{\kappa x} + a_2 e^{-\kappa x}$$

2. If $\lambda = 0$, we obtain

$$\psi(x) = a_3 x + a_4$$

3. If $\lambda < 0$, we can set $\lambda = -\kappa^2$ and obtain

$$\psi(x) = a_5 \cos \kappa x + a_6 \sin \kappa x$$

Applying Boundary Conditions

$$y(0, t) = y(L, t) = 0 \rightarrow \psi(0) = \psi(L) = 0$$

1. If $\lambda > 0$, $\psi(x) = a_1 e^{\kappa x} + a_2 e^{-\kappa x}$

$$\psi(0) = a_1 + a_2 = 0, \quad \psi(L) = a_1 e^{\kappa L} + a_2 e^{-\kappa L} = 0$$

$$\therefore a_1 = a_2 = 0$$

2. If $\lambda = 0$, $\psi(x) = a_3 x + a_4$

$$\psi(0) = a_4 = 0, \quad \psi(L) = a_3 L + a_4 = 0$$

$$\therefore a_3 = a_4 = 0$$

Applying Boundary Conditions

$$y(0, t) = y(L, t) = 0 \rightarrow \psi(0) = \psi(L) = 0$$

3. If $\lambda < 0$, $\psi(x) = a_5 \cos \kappa x + a_6 \sin \kappa x$

$$\psi(0) = a_5 = 0, \quad \psi(L) = a_5 \cos \kappa L + a_6 \sin \kappa L = 0$$

$$\therefore a_5 = 0 \text{ and } \kappa = n\pi/L$$

The only non-zero solution is

$$\psi(x) = a_6 \sin\left(\frac{n\pi x}{L}\right)$$

when $\lambda = -\kappa^2$ and $\kappa = n\pi/L$

Temporal Part

$$\frac{1}{c^2 \theta} \frac{d^2 \theta}{dt^2} = \frac{1}{\psi} \frac{d^2 \psi}{dx^2} = \lambda = -\kappa^2 = -\left(\frac{n\pi}{L}\right)^2$$

$$\frac{d^2 \theta}{dt^2} = -\left(\frac{n\pi}{L}\right)^2 c^2 \theta$$

Since the separation constant is negative,

$$\theta(t) = b_1 \cos\left(\frac{n\pi c}{L} t\right) + b_2 \sin\left(\frac{n\pi c}{L} t\right)$$

Applying Initial Conditions

$$\theta(t) = b_1 \cos\left(\frac{n\pi c}{L} t\right) + b_2 \sin\left(\frac{n\pi c}{L} t\right)$$

$$y(x, 0) = 0 \rightarrow \theta(0) = 0$$

Substitution leads to

$$\theta(0) = b_1 = 0$$

Hence,

$$\theta(t) = b_2 \sin\left(\frac{n\pi c}{L} t\right)$$

Total Solution

- Spatial solution

$$\psi(x) = a_6 \sin\left(\frac{n\pi x}{L}\right)$$

- Temporal solution

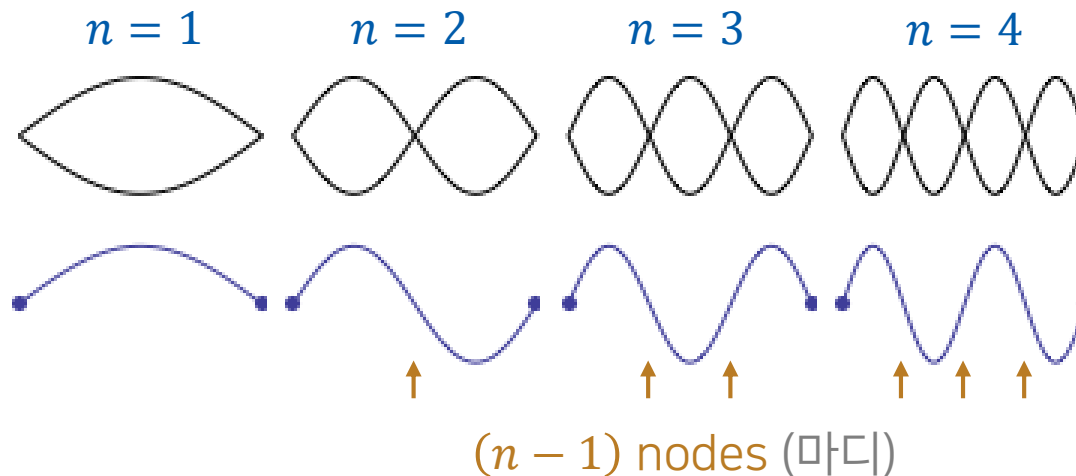
$$\theta(t) = b_2 \sin\left(\frac{n\pi ct}{L}\right)$$

- Total solution

$$y(x, t) = \psi(x)\theta(t) = A \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi ct}{L}\right)$$

Standing Waves (정상파)

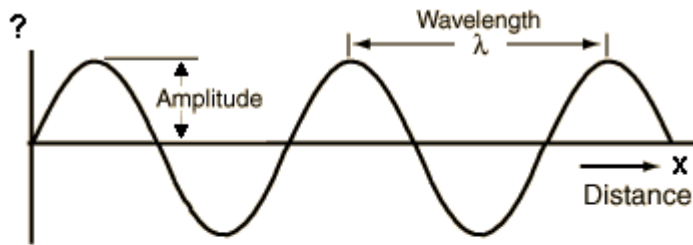
$$y(x, t) = \psi(x)\theta(t) = A \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi ct}{L}\right)$$



(Image: <https://www.acs.psu.edu/drussell/demos/standingwaves/standingwaves.html>)

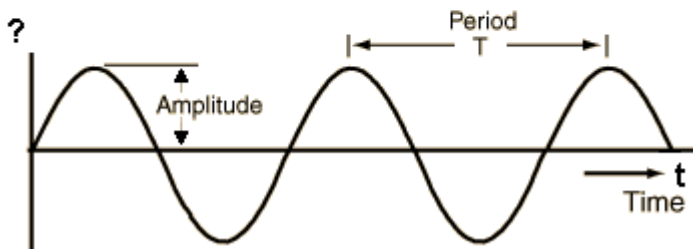
Wavelength and Period (파장과 주기)

$$y(x, t) = \psi(x)\theta(t) = A \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi ct}{L}\right)$$



- Wavelength λ

$$n\pi\lambda/L = 2\pi \rightarrow \lambda = 2L/n$$



- Period τ

$$n\pi c\tau/L = 2\pi \rightarrow \tau = 2L/nc$$

The Schrödinger Equation

- 1-dimensional time-dependent Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + \mathcal{V}(x, t)\Psi$$

- $\Psi = \Psi(x, t)$: wave function
- m : mass of a particle
- $\mathcal{V}(x, t)$: potential energy
- $\hbar = h/2\pi$, where h is Planck's constant
- Consider the case $\mathcal{V}(x, t) = \mathcal{V}(x)$!

Separation of Variables

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + \mathcal{V}(x)\Psi$$

Assume $\Psi(x, t) = \psi(x)\theta(t)$:

$$i\hbar \psi(x)\theta'(t) = -\frac{\hbar^2}{2m} \psi''(x)\theta(t) + \mathcal{V}(x)\psi(x)\theta(t)$$

Dividing both sides by $\psi(x)\theta(t)$,

$$\underbrace{i\hbar \frac{\theta'(t)}{\theta(t)}}_{\text{depends only on } t} = \underbrace{-\frac{\hbar^2}{2m} \frac{\psi''(x)}{\psi(x)} + \mathcal{V}(x)}_{\text{depends only on } x}$$

Separation Constant

$$i\hbar \frac{\theta'(t)}{\theta(t)} = -\frac{\hbar^2}{2m} \frac{\psi''(x)}{\psi(x)} + \mathcal{V}(x) = E$$

- Spatial part (time-independent Schrödinger eqn.)

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \mathcal{V}(x)\psi(x) = E\psi(x)$$

- Temporal part

$$i\hbar \frac{d\theta}{dt} = E\theta(t)$$

$$\theta(t) = e^{-iEt/\hbar}$$